25901933 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCCSC201 - DATA STRUCTURES USING C

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. What is a parameterized function? [K] / [CO1]
2. Discuss how does a stack help in postfix expression evaluation? [U] / [CO2]
3. Examine binary tree commmonly used to implement heaps. [An] / [CO3]
4. Explain why merge sort is considered a divide and conquer algorithm. [U] / [CO1]
5. List the steps for deleting the first node of a singly linked list. [K] / [CO3]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **5** marks

6. Interpret how the number of comparisons in linear search and binary search changes as the dataset size increases. [U] / [CO1]

7. Describe the array representation of a queue. How are the ‘front’ and ‘rear’ pointers managed? [U] / [CO2]

8. Compare malloc(), calloc(), and realloc() with examples. [An] / [CO3]

9. Demonstrate the use of recursion to find factorial of a number. [U] / [CO1]

10. Illustrate the step-by-step changes in a stack (array representation) when the following operations are performed: push(10), push(20), pop() and push(30). [A] / [CO2]

11. Analyse the structural and functional differences between a binary tree and a binary search tree. Use an example of a binary search tree to justify your analysis. [An] / [CO3]

12. Explain how elements are shifted during insertion and deletion in a linear aray. [U] / [CO1]

[5x5 = 25]

Part C

Essay Questions

Answer any **1** questions

Each question carries **15** marks

13. Define data structures in C and list different types with examples. [K] / [CO1]

14. Define circular queue and list its basic operations. [K] / [CO2]

[15x1 = 15]